Inequalities in Two Variables

AQA · KS4 Year 9–10

Answer each question in your workbook. Draw graphs carefully on squared paper.

Method

Draw the boundary line (sign =)   Dashed for </>, solid for /   Test a point not on the line

Shade the side that works   Several inequalities shade the overlap

Common mistakes

- Solid line for < or > (should be dashed).
- Dashed line for or (should be solid).
- Shading the wrong side — always test a point.
- Region must satisfy ALL the inequalities.

Worked example

Show y x + 2.

Draw y = x + 2 SOLID ( ). Test (0,0): 0 2 true shade the side with (0,0).

That shaded area below the line is the region.

Visual aid

Visual Example

On x–y axes, each inequality gives a half-plane: a boundary line (dashed for strict, solid for inclusive) divides the grid in two, and you shade the side where the inequality holds. The region for several inequalities is where all the shadings overlap.

Questions

Answer each question in your workbook. Draw graphs carefully on squared paper.

Q1 of 8 **[2]** ●○○○○

Solid or dashed line? a) y > 2x b) y   x + 1

Q2 of 8 **[2]** ●○○○○

Satisfies the inequality? a) (1,4) in y > x + 2? b) (2,1) in y   2x - 1?

Q3 of 8 **[3]** ●●○○○

Draw the boundary and shade the region: a) x > 2 b) y   3 c) y < x

Q4 of 8 **[3]** ●●○○○

Does (0,0) lie in each region? a) y < x + 1 b) y   2x - 3 c) y > x

Q5 of 8 **[4]** ●●●○○

Shade R satisfying x   0, y   0, x + y   4. State the three corners.

Q6 of 8 **[3]** ●●●○○

A region is above y = 1, right of x = 1, below y = 5 - x. Write the three inequalities.

Q7 of 8 **[4]** ●●●●○

R: y   1, y   2x - 1, x   4. a) Which lines are solid? b) Is (3, 2) inside R? Show working.

Q8 of 8 **[5]** ●●●●●

Van: x small + y large parcels, x + y   10, x   2, y   3. a) Meaning of each. b) A valid (x,y). c) Greatest total and an (x,y) achieving it.